Midterm Project

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Concept + Methodology

The concept for this "piece" (and, more accurately, the messy collection of associated python and bash scripts) stems from an interest in both the serial music concerned with pitch classes and tone rows (e.g. Boulez and Webern) and the algorithmic music concerned with structure and mathematical generation (e.g. Stockhausen and Xenakis). Thus, this piece is composed of two musical segments to be understood in counterpoint with another. Both segments are generated the same way:

1. First, a tone row and corresponding tone matrix is generated.
2. Second, a sequence of Rows from said matrix is generated.
3. Third, a sequence of pitches is generated using the following concatenation rule:

   > Given two sequences of pitches, the second sequence is appended to the first at the last occurrence of the first pitch in the second sequence. Thus, joining [0 1 2 3 4 5] and [3 2 1] produces [0 1 2 3 2 1].

4. Finally, a sequence of durations is generated randomly exclusively for aural interest. The sequence of durations serves no structural purpose and is simply used for aural scaffolding to support the listener.

Additionally, the metadata (that is, the sequence of rows and their transitions) is annotated underneath the music. The first segment, which shall be referred to as the 'flat' segment, is subject to the following constraints:

- A minimum of sixteen ‘measures’ of music, guaranteed by a minimum of sixty-four generated pitches.
- There is no preference or weighting given to the sequence of rows selected; each possible subsequent row is equally likely, and the probability distribution looks ‘flat’ - hence the name.
- The sequence of rows ends on `P0` or `I0`.

The second segment, uses a Markov process to generate the sequence of rows and is subject to the following constraints:
replacewithGenerated Music

- Again, by the same methods, a minimum of sixteen ‘measures’ and sixty-four pitches is generated.
- At each step in the sequence of rows, the subsequent row is chosen by a Markov process detailled by the Markov table in the appropriate section.
- The sequence of rows ends on `P0` or `I0`.

This second segment is appropriately named the ‘Markov’ segment.

Generated Music

Tone Row

The Generated Tone Row is:

Tone Matrix

The table generated for the tone matrix is:

1	9	0	8	7	3	5	6	t	2	4	e
5	1	4	0	e	7	9	t	2	6	8	3
2	t	1	9	8	4	6	7	e	3	5	0
6	2	5	1	0	8	t	e	3	7	9	4
7	3	6	2	1	9	e	0	4	8	t	5
e	7	t	6	5	1	3	4	8	0	2	9
9	5	8	4	3	e	1	2	6	t	0	7
8	4	7	3	2	t	0	1	5	9	e	6
4	0	3	e	t	6	8	9	1	5	7	2
0	8	e	7	6	2	4	5	9	1	3	t
t	6	9	5	4	0	2	3	7	e	1	8
3	e	2	t	9	5	7	8	0	4	6	1

‘Flat’ Segment

The music generated with the aforementioned parameters is displayed below:

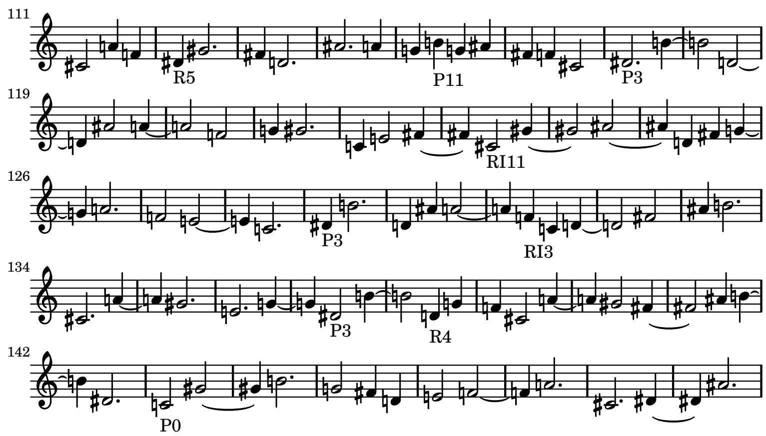
Markov Chain Table

The table generated for the Markov Chain is:

x	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	0.0	0.0	0.0	0.1	0.03	0.0	0.0	0.0	0.01	0.0	0.08	0.01	0.0	0.0	0.03
2	0.06	0.0	0.0	0.1	0.03	0.0	0.0	0.0	0.0	0.0	0.01	0.02	0.0	0.0	0.0
3	0.07	0.0	0.0	0.07	0.05	0.0	0.0	0.0	0.0	0.0	0.06	0.0	0.0	0.0	0.04
4	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
5	0.04	0.0	0.0	0.11	0.0	0.0	0.0	0.0	0.01	0.0	0.06	0.0	0.0	0.0	0.07
6	0.0	0.0	0.0	0.1	0.04	0.0	0.0	0.0	0.02	0.0	0.04	0.0	0.0	0.0	0.06
7	0.03	0.0	0.0	0.05	0.05	0.0	0.0	0.0	0.03	0.01	0.07	0.0	0.0	0.0	0.0
8	0.02	0.0	0.0	0.11	0.0	0.0	0.0	0.0	0.0	0.03	0.09	0.0	0.0	0.0	0.0
9	0.05	0.0	0.0	0.03	0.02	0.0	0.0	0.0	0.0	0.0	0.09	0.0	0.0	0.0	0.07
10	0.05	0.0	0.0	0.08	0.0	0.0	0.0	0.01	0.03	0.0	0.03	0.01	0.0	0.0	0.05
11	0.07	0.0	0.0	0.08	0.01	0.0	0.0	0.0	0.04	0.02	0.0	0.0	0.0	0.0	0.05
12	0.0	0.0	0.0	0.11	0.0	0.0	0.0	0.0	0.0	0.03	0.09	0.0	0.0	0.0	0.02
13	0.04	0.0	0.0	0.11	0.0	0.0	0.0	0.0	0.01	0.0	0.06	0.0	0.0	0.0	0.07
14	0.0	0.0	0.0	0.1	0.04	0.0	0.0	0.0	0.02	0.0	0.04	0.0	0.0	0.0	0.06
15	0.03	0.0	0.0	0.05	0.05	0.0	0.0	0.0	0.03	0.01	0.07	0.0	0.0	0.0	0.0
16	0.02	0.0	0.0	0.11	0.0	0.0	0.0	0.0	0.0	0.03	0.09	0.0	0.0	0.0	0.0
17	0.05	0.0	0.0	0.03	0.02	0.0	0.0	0.0	0.0	0.0	0.09	0.0	0.0	0.0	0.07
18	0.05	0.0	0.0	0.08	0.0	0.0	0.0	0.01	0.03	0.0	0.03	0.01	0.0	0.0	0.05
19	0.07	0.0	0.0	0.08	0.01	0.0	0.0	0.0	0.04	0.02	0.0	0.0	0.0	0.0	0.05
20	0.0	0.0	0.0	0.11	0.0	0.0	0.0	0.0	0.0	0.03	0.09	0.0	0.0	0.0	0.02
21	0.0	0.0	0.0	0.1	0.03	0.0	0.0	0.0	0.01	0.0	0.08	0.01	0.0	0.0	0.03
22	0.06	0.0	0.0	0.1	0.03	0.0	0.0	0.0	0.0	0.0	0.01	0.02	0.0	0.0	0.0
23	0.07	0.0	0.0	0.07	0.05	0.0	0.0	0.0	0.0	0.0	0.06	0.0	0.0	0.0	0.04
24	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
25	0.07	0.0	0.0	0.07	0.05	0.0	0.0	0.0	0.0	0.0	0.06	0.0	0.0	0.0	0.04
26	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
27	0.04	0.0	0.0	0.11	0.0	0.0	0.0	0.0	0.01	0.0	0.06	0.0	0.0	0.0	0.07
28	0.0	0.0	0.0	0.1	0.04	0.0	0.0	0.0	0.02	0.0	0.04	0.0	0.0	0.0	0.06
29	0.03	0.0	0.0	0.05	0.05	0.0	0.0	0.0	0.03	0.01	0.07	0.0	0.0	0.0	0.0
30	0.02	0.0	0.0	0.11	0.0	0.0	0.0	0.0	0.0	0.03	0.09	0.0	0.0	0.0	0.0
31	0.05	0.0	0.0	0.03	0.02	0.0	0.0	0.0	0.0	0.0	0.09	0.0	0.0	0.0	0.07
32	0.05	0.0	0.0	0.08	0.0	0.0	0.0	0.01	0.03	0.0	0.03	0.01	0.0	0.0	0.05
33	0.07	0.0	0.0	0.08	0.01	0.0	0.0	0.0	0.04	0.02	0.0	0.0	0.0	0.0	0.05
34	0.0	0.0	0.0	0.11	0.0	0.0	0.0	0.0	0.0	0.03	0.09	0.0	0.0	0.0	0.02
35	0.0	0.0	0.0	0.1	0.03	0.0	0.0	0.0	0.01	0.0	0.08	0.01	0.0	0.0	0.03
36	0.06	0.0	0.0	0.1	0.03	0.0	0.0	0.0	0.0	0.0	0.01	0.02	0.0	0.0	0.0
37	0.07	0.0	0.0	0.07	0.05	0.0	0.0	0.0	0.0	0.0	0.06	0.0	0.0	0.0	0.04
38	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
39	0.04	0.0	0.0	0.11	0.0	0.0	0.0	0.0	0.01	0.0	0.06	0.0	0.0	0.0	0.07
40	0.0	0.0	0.0	0.1	0.04	0.0	0.0	0.0	0.02	0.0	0.04	0.0	0.0	0.0	0.06
41	0.03	0.0	0.0	0.05	0.05	0.0	0.0	0.0	0.03	0.01	0.07	0.0	0.0	0.0	0.0
42	0.02	0.0	0.0	0.11	0.0	0.0	0.0	0.0	0.0	0.03	0.09	0.0	0.0	0.0	0.0
43	0.05	0.0	0.0	0.03	0.02	0.0	0.0	0.0	0.0	0.0	0.09	0.0	0.0	0.0	0.07
44	0.05	0.0	0.0	0.08	0.0	0.0	0.0	0.01	0.03	0.0	0.03	0.01	0.0	0.0	0.05
45	0.07	0.0	0.0	0.08	0.01	0.0	0.0	0.0	0.04	0.02	0.0	0.0	0.0	0.0	0.05
46	0.0	0.0	0.0	0.11	0.0	0.0	0.0	0.0	0.0	0.03	0.09	0.0	0.0	0.0	0.02
47	0.0	0.0	0.0	0.1	0.03	0.0	0.0	0.0	0.01	0.0	0.08	0.01	0.0	0.0	0.03
48	0.06	0.0	0.0	0.1	0.03	0.0	0.0	0.0	0.0	0.0	0.01	0.02	0.0	0.0	0.0

'Markov' Segment

The music generated by using the above Markov Table in a Markov Chain Process is displayed below:


111
R5
P11
P3
119
RI11
126
P3
RI3
134
P3
R4
142
P0

Computer Code

The music is generated by the `midterm.py` script in the `./python` folder.

To generate a new `midterm.pdf` document, run the script `./generate.sh`.

The generated document is available as `assignment-print.pdf` in the root directory.